

Week 3 Discussion: Futures (Lec 5–6) Review

Main focus: Futures basics, daily settlement & margins, hedging & basis risk. (Quiz: quick reminders only.)

1 Rapid Review: Lec 5–6 Essentials

1) Futures vs Forward + Close-out P&L

- **Forward (OTC):** typically **no daily settlement**; credit exposure to counterparty; often harder to close out (needs renegotiation/offsetting OTC contract).
- **Futures (exchange + clearinghouse):** **daily settlement (mark-to-market)** via margin; can **close out** anytime by taking the **opposite** position; credit risk is mitigated by margin + clearinghouse.

Close-out P&L over $[t_1, t_2]$:

$$\text{Long futures: P\&L} = F_{t_2} - F_{t_1}, \quad \text{Short futures: P\&L} = F_{t_1} - F_{t_2}.$$

Daily settlement changes the timing of cashflows (realized daily), not the algebraic total P&L over the horizon.

2) Margins & Daily Settlement

Margin account: collateral posted with broker for a futures position.

- **Initial Margin (IM):** deposit when position is opened.
- **Maintenance Margin (MM):** threshold; if balance < MM $\Rightarrow$ **margin call**.
- **Variation Margin:** deposit/withdraw due to daily P&L; after a margin call, deposit to restore balance back to IM (typical convention).

Daily P&L (contract multiplier m):

$$\Delta \text{P\&L}_{\text{long}} = m(F_t - F_{t-1}), \quad \Delta \text{P\&L}_{\text{short}} = m(F_{t-1} - F_t).$$

Margin intuition (order-of-magnitude):

$$\mathbb{E}(|F_j - F_{j-1}|) \ll \text{Initial Margin} \ll F_T.$$

2 Practice Problems

P1. Margin call (silver futures).

Suppose that you enter into a **short** futures contract to sell July silver for \$17.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is \$4,000, and the maintenance margin is \$3,000.

- (a) What change in the futures price will lead to a margin call?

- (b) What happens if you do not meet the margin call?

P2. Daily settlement across tax year (crude oil futures).

Suppose that in September 2021 a company takes a **long** position in a contract on May 2022 crude oil futures. It closes out its position in March 2022. The futures price (per barrel) is \$48.30 when it enters into the contract, \$49.10 at the end of December 2021, and \$50.50 when it closes out its position. One contract is for the delivery of 1,000 barrels. The company has a December 31 year end.

- (a) What is the company's **total profit**?

- (b) **When** is the profit **realized** (cashflow timing)? *Hint: daily settlement.*

Rapid-review checkpoints (common mistakes)

- **Sign check:** long futures gains when F increases; short futures loses when F increases.
- **Daily settlement:** does *not* change total close-out P&L, but it *does* change **when** cash gains/losses hit the margin account.
- **Margin call:** triggered by **margin balance** falling below MM (convert price move $\rightarrow$ dollars using contract size).